

GeoRisk-SPC: Geometry-conditioned Pseudo-label Risk Supervision for Semi-supervised Surgical Instrument Segmentation

Author Names

Abstract—Semi-supervised surgical instrument segmentation reduces annotation cost, but pseudo-label errors concentrate at thin tool boundaries, occlusion interfaces, and depth discontinuities. Existing methods strengthen pseudo-label generation or consistency regularization but still treat confidence as the main trust criterion. We propose GeoRisk-SPC, a geometry-conditioned pseudo-label risk supervision framework that uses depth discontinuity to decide where pseudo-labels should not be trusted. GeoRisk-SPC builds a pixel-wise risk map from monocular depth discontinuity, teacher uncertainty, and geometry-semantic conflict; applies hard pseudo-label loss to low-risk pixels; and applies soft structural consistency with risk-guided feature perturbation to high-risk pixels. The GeoRisk-SPC-DG variant adds DepthGuiderV4, a gated multi-scale depth encoder with residual modulation. On EndoVis 2017 with 4-fold cross-validation, GeoRisk-SPC-DG achieves test Dice of 88.26%, 90.42%, 90.90%, and 91.41% at 5–40% labels, outperforming baselines at 10–40%. At 20% labels, Boundary F1 reaches 92.42% versus 89.66% for Mean Teacher. The depth encoder is the most reliable contributor (+1.40 points over MT). Risk localization identifies unreliable pseudo-label regions with a large accuracy gap between low-risk and high-risk pixels, though fixed-threshold calibration remains a limitation. These findings support depth both as a feature source and as a pseudo-label risk prior.

Index Terms—semi-supervised medical image segmentation, surgical instrument segmentation, pseudo-label reliability, geometric risk localization, depth-guided segmentation

I. INTRODUCTION

Surgical instrument segmentation is a core perception problem for robotic-assisted minimally invasive surgery. Accurate masks support tool tracking, workflow analysis, augmented reality guidance, human-robot interaction, and downstream surgical scene understanding [1], [2], [3]. Despite strong progress in deep segmentation, surgical video remains difficult to annotate at scale. Instrument shafts are thin, tips are small, metallic surfaces are reflective, and boundaries are frequently obscured by tissue, smoke, blood, motion blur, or another tool. Dense masks therefore require expert labor and careful quality control.

Semi-supervised learning (SSL) is attractive because unlabeled surgical frames are abundant. The dominant SSL recipe combines pseudo-labeling and consistency regularization: a teacher, an exponential moving average model, or a peer network predicts targets on weakly perturbed unlabeled images, and a student is trained under stronger perturbations [4], [5], [6], [7], [8]. The 2024–2026 literature has intensified this

direction. Generic semi-supervised segmentation has advanced through correlation-based propagation, boundary-aware confidence weighting, and stronger weak-to-strong formulations [9], [10], [11]. Medical SSL has moved toward cross-style consistency, worst-case-aware decoupled training, adversarial reinforcement learning, probabilistic prototypes, and collaborative segmentation paradigms [12], [13], [14], [15], [16]. Surgical SSL has also become more competitive through uncertainty-guided transformer supervision and adversarial pseudo-label augmentation [17], [18].

These methods make pseudo-labels stronger, but surgical scenes expose a remaining failure mode: pseudo-label reliability is spatially heterogeneous. A teacher can be confident and correct in a homogeneous tool shaft while being wrong at the adjacent tool edge, tip, reflection, or occlusion boundary. A global confidence threshold does not fully capture this structure. Hard pseudo-label supervision on unreliable boundary pixels can reinforce foreground leakage and fragmented instruments, whereas uniformly suppressing unlabeled supervision wastes reliable interior and background pixels.

Our key observation is that unreliable pseudo-label regions are geometrically structured. Monocular depth prediction is now a practical dense cue [19], and RGB-D segmentation studies consistently show that depth helps with object layout and boundaries [20], [21], [22]. In endoscopic images, instrument-tissue interfaces and tool overlap often create local depth discontinuities. These discontinuities are not a substitute for semantic segmentation labels, but they are useful for judging pseudo-label risk. This leads to the sharp design principle of GeoRisk-SPC: depth is not used as another segmentation modality only; it is used to decide where pseudo-labels should not be trusted.

We propose GeoRisk-SPC (Geometry-conditioned Pseudo-label Risk Supervision), a semi-supervised framework for surgical instrument segmentation. GeoRisk-SPC estimates a risk map by combining local depth discontinuity, teacher entropy, and geometry-semantic boundary conflict. It then partitions unlabeled pixels into low-risk high-confidence regions and high-risk ambiguous regions. Low-risk pixels receive hard pseudo-label supervision. High-risk pixels avoid hard teacher labels and instead receive soft structural consistency between clean and risk-perturbed student branches. The full GeoRisk-SPC-DG model adds a depth-guided geometric encoder, DepthGuiderV4, which injects multi-scale depth and depth-gradient features into RGB features through gated residual modulation.

Our contributions are:

- We formulate semi-supervised surgical instrument segmentation as a geometry-conditioned pseudo-label risk problem, shifting the question from how to generate a stronger pseudo-label everywhere to where a pseudo-label should be trusted.
- We propose a risk localization module that combines monocular depth discontinuity, teacher uncertainty, and geometry-semantic conflict to select supervision types for unlabeled pixels.
- We design region-aware supervision with hard pseudo-label learning in low-risk pixels and soft structural consistency in high-risk pixels, together with risk-guided feature perturbation.
- We introduce DepthGuiderV4, a gated multi-scale depth encoder that provides the strongest empirical contribution in our study.
- We report 4-fold test results on EndoVis 2017 at 5%, 10%, 20%, and 40% label ratios, and we explicitly analyze the non-monotonic ablation behavior caused by fixed-threshold risk calibration.

II. RELATED WORK

A. Surgical Instrument Segmentation

Surgical instrument segmentation has been extensively studied through EndoVis challenge datasets and surgical data science benchmarks [1], [2], [3]. Recent work addresses annotation efficiency and realism through image-to-image translation, synthetic image generation, and active learning [23], [24], [25]. The task differs from natural-scene segmentation because instrument boundaries are thin, reflective, articulated, and clinically important. GeoRisk-SPC targets this low-label regime with a semi-supervised method specialized for boundary and geometry-dependent pseudo-label risk.

B. Semi-supervised Semantic Segmentation

Mean Teacher stabilizes consistency targets through exponential moving average weights [4]. CPS uses two networks for cross pseudo supervision [6]. U2PL explicitly exploits unreliable pseudo-labels as negative samples for contrastive learning [7]. UniMatch revisits weak-to-strong consistency with unified pseudo-label training [8], while UniMatch V2 further pushes generic semi-supervised segmentation performance [11]. CorrMatch propagates labels through correlation matching [9], and CW-BASS introduces confidence-weighted boundary-aware learning [10]. These methods are strong baselines, but their reliability criteria are mainly confidence, augmentation, correlation, or class-wise thresholding. GeoRisk-SPC instead introduces depth discontinuity as an external spatial prior for pseudo-label risk.

C. Semi-supervised Medical and Surgical Segmentation

Medical SSL segmentation has evolved from uncertainty-aware self-ensembling [26] and uncertainty-rectified pyramid consistency [27], [28] to stronger recent TMI and MIC-CAI methods. Cross-style consistency combines shape-aware

and local context constraints [12]. Decoupled training considers worst-case learning [13]. Generative adversarial reinforcement learning targets low-contrast and small-object segmentation [14]. Probabilistic representations and prototype learning model semi-supervised uncertainty [15], and Segment Together proposes a versatile collaborative paradigm [16]. In surgical instrument segmentation, MMS uses min-max similarity for contrastive semi-supervised learning [29], Surgivisor applies uncertainty-guided transformer supervision [17], and SegMatch uses adversarial augmentation for pseudo-label learning [18]. These works confirm the competitiveness of SSL under limited labels. Our method is complementary: it localizes where pseudo-labels are risky before deciding which supervision loss to apply.

D. Depth-aware Segmentation and Pseudo-label Quality

Depth-adaptive networks and depth-aware attention demonstrate that depth cues can improve semantic segmentation by emphasizing geometry and boundaries [20], [21]. DPT-style monocular dense prediction makes depth maps available without an RGB-D sensor [19], and recent RGB-D representation learning continues to improve fusion architectures [22]. Most depth-aware segmentation methods use depth as an additional feature source. GeoRisk-SPC uses depth in a different role: depth discontinuity estimates pseudo-label risk on unlabeled data. This connects depth-aware segmentation with pseudo-label quality control, where classical pseudo-labeling, MixMatch, FlexMatch, uncertainty-aware learning, and correlation propagation seek more reliable targets [30], [5], [31], [26], [9]. The missing piece is spatial trust: a pseudo-label may be reliable in one pixel and harmful in the neighboring boundary pixel. GeoRisk-SPC directly models this spatial trust.

III. METHOD

A. Problem Formulation and Overview

Let $\mathcal{D}_l = \{(x_i^l, y_i^l, d_i^l)\}_{i=1}^{N_l}$ be labeled RGB images, binary masks, and pre-computed monocular depth maps, and let $\mathcal{D}_u = \{(x_i^u, d_i^u)\}_{i=1}^{N_u}$ be unlabeled RGB-depth pairs. GeoRisk-SPC follows an EMA teacher-student framework. The student S_ϕ is optimized by supervised and unlabeled losses, while the teacher T_θ is updated by $\theta \leftarrow \alpha\theta + (1 - \alpha)\phi$.

The main design is region-aware unlabeled supervision. The teacher produces probabilities p_t on unlabeled images. A geometry-conditioned risk map R partitions pixels into high-risk pixels M_r and low-risk high-confidence pixels M_l . Low-risk pixels receive hard pseudo-label cross-entropy. High-risk pixels receive soft consistency between the student's clean branch and a perturbed branch. GeoRisk-SPC-UNet uses RGB-D input with a UNet-style [32] dual decoder. GeoRisk-SPC-DG replaces the encoder with DepthGuiderV4. Algorithm 1 summarizes training.

B. Geometry-aware Risk Localization

For an unlabeled image with depth d_u , GeoRisk-SPC first computes a locally normalized depth map:

$$d_{rel} = \frac{d_u - \mu_{local}(d_u)}{\sigma_{local}(d_u) + \epsilon}, \quad (1)$$

Algorithm 1 GeoRisk-SPC Training

Require: Labeled batch (x_l, y_l, d_l) , unlabeled batch (x_u, d_u) , student S_ϕ , EMA teacher T_θ

- 1: Compute teacher probabilities $p_t = T_\theta(x_u, d_u)$
- 2: Compute risk map R from d_u and p_t
- 3: Build $M_r = \mathbb{I}[R > \tau_r]$ and $M_l = \mathbb{I}[\max_c p_t^c > \tau_c](1 - M_r)$
- 4: Apply additive-noise student perturbation to unlabeled inputs
- 5: Predict clean and risk-perturbed student outputs
- 6: Compute $\mathcal{L}_{sup}$, $\mathcal{L}_{pl}$, $\mathcal{L}_{cons}$, and $\mathcal{L}_{bd}$
- 7: Update ϕ by minimizing $\mathcal{L}_{sup} + w(t)(\lambda_{pl}\mathcal{L}_{pl} + \lambda_{cons}\mathcal{L}_{cons} + \lambda_{bd}\mathcal{L}_{bd})$
- 8: Update $\theta \leftarrow \alpha\theta + (1 - \alpha)\phi$

where μ_{local} and σ_{local} are computed in a local window. Depth discontinuity is the Sobel [33] gradient magnitude:

$$G_d = \|S * d_{rel}\|. \quad (2)$$

Teacher uncertainty is Shannon entropy [34]:

$$U_t = - \sum_{c=1}^C p_t^c \log(p_t^c + \epsilon). \quad (3)$$

The prediction boundary map is the average class-probability gradient:

$$B_p = \frac{1}{C} \sum_{c=1}^C \|S * p_t^c\|. \quad (4)$$

The implementation uses batch-global min-max normalization for each map:

$$\bar{A} = \frac{A - \min_{b,h,w} A}{\max_{b,h,w} A - \min_{b,h,w} A + \epsilon}. \quad (5)$$

This normalization is applied over the full tensor passed to the risk module, not independently for every image. Geometry-semantic conflict is:

$$C_{conf} = |\bar{G}_d - \bar{B}_p|. \quad (6)$$

The full risk map is:

$$R = \bar{U}_t \bar{G}_d + \lambda_c C_{conf}. \quad (7)$$

The masks are:

$$M_r = \mathbb{I}[R > \tau_r], \quad (8)$$

$$M_l = \mathbb{I}[\max_c p_t^c > \tau_c](1 - M_r). \quad (9)$$

By construction, M_l is the hard pseudo-label region and M_r is the region where hard labels are avoided.

C. Region-aware Supervision

The labeled loss combines Dice and cross-entropy:

$$\mathcal{L}_{sup} = \frac{1}{2}(\mathcal{L}_{dice} + \mathcal{L}_{ce}). \quad (10)$$

Let $\hat{y} = \arg \max_c p_t^c$. Low-risk pixels receive hard pseudo-label supervision:

$$\mathcal{L}_{pl} = \frac{\sum_{i,j} M_l^{(i,j)} \text{CE}(z_{clean}^{(i,j)}, \hat{y}^{(i,j)})}{\sum_{i,j} M_l^{(i,j)} + \epsilon}. \quad (11)$$

High-risk pixels receive soft consistency. The implementation uses PyTorch `KLDivLoss(log_pert, clean.detach())`, which corresponds to the Kullback-Leibler divergence [35]:

$$\mathcal{L}_{cons} = \frac{\sum_{i,j} M_r^{(i,j)} D_{KL}(\text{sg}[p_{clean}^{(i,j)}] \| p_{pert}^{(i,j)})}{\sum_{i,j} M_r^{(i,j)} + \epsilon}. \quad (12)$$

The clean branch is detached and the perturbed branch is optimized. Boundary consistency compares Sobel gradients of the clean and perturbed probabilities:

$$\mathcal{L}_{bd} = \frac{\sum_{i,j} M_r^{(i,j)} \|\|S * p_{clean} - \|S * p_{pert}\|\|}{\sum_{i,j} M_r^{(i,j)} + \epsilon}. \quad (13)$$

The total loss is:

$$\mathcal{L} = \mathcal{L}_{sup} + w(t)(\lambda_{pl}\mathcal{L}_{pl} + \lambda_{cons}\mathcal{L}_{cons} + \lambda_{bd}\mathcal{L}_{bd}), \quad (14)$$

where $w(t)$ is a sigmoid ramp-up schedule.

D. Structural Perturbation Consistency

GeoRisk-SPC perturbs bottleneck features only in high-risk regions. The risk mask is bilinearly resized to the bottleneck resolution. For channel dropout mask $D \in \{0, 1\}^{B \times C \times 1 \times 1}$ and Gaussian noise $\eta \sim \mathcal{N}(0, \sigma^2)$, the implemented perturbation is:

$$f_{pert} = f_{clean} \odot (1 - \hat{M}_r \odot (1 - D)) + \hat{M}_r \odot \eta. \quad (15)$$

The perturbation is applied only to the perturbed decoder branch. At inference, the model uses only the clean branch, so this module does not add test-time cost.

E. Depth-guided Geometric Encoder

DepthGuiderV4 injects depth information into every UNet encoder level. Given RGB feature f^l and resized depth d^l , the module computes depth-gradient geometry:

$$g^l = [\nabla_x d^l, \nabla_y d^l, \|\nabla d^l\|]. \quad (16)$$

Separate convolutional encoders produce depth and geometry features, which are mixed by a depth mixer. Local, pooled, and global depth contexts are built from the mixed feature. A scale router predicts softmax weights over the three contexts, and the weighted context interacts with projected RGB features through multiplication and absolute-difference terms. The fused feature is passed through a sigmoid gate and projected as a residual:

$$q^l = F^l([u^l, c^l, u^l \odot \sigma(c^l), |u^l - c^l|]), \quad (17)$$

$$f_{out}^l = f^l + \sigma(G^l([q^l, c^l])) \odot P^l(q^l), \quad (18)$$

where u^l is the projected RGB feature, c^l is the routed depth context, and q^l is the fused interaction feature. The gate and

residual projection are initialized to zero, so the module starts as an identity mapping and learns depth injection only when useful. This encoder is distinct from risk localization: the encoder uses depth to improve representation, whereas the risk module uses depth discontinuity to decide pseudo-label trust.

IV. EXPERIMENTS

A. Dataset and Protocol

We evaluate on EndoVis 2017 Task 1 [2], a binary instrument segmentation benchmark with 1800 images from 10 sequences. We use 4-fold sequence-based cross-validation and report test Dice under 5%, 10%, 20%, and 40% labeled training data. Depth maps are pre-computed from monocular depth estimation [19]. All reported main results are test-set Dice averaged over folds; they are not validation-best Dice.

B. Implementation Details

All methods use the same image resolution, UNet backbone, and 4-fold protocol. GeoRisk-SPC uses Adam with learning rate 10^{-4} and weight decay 10^{-4} , batch size 24 with 12 labeled and 12 unlabeled samples, 30,000 maximum iterations, early stopping patience of 9,000 iterations, EMA decay 0.99, and a UNet family backbone with 16 initial filters. The default risk parameters are $\tau_r = 0.5$, $\tau_c = 0.9$, and $\lambda_c = 0.5$. Loss weights are $\lambda_{pl} = 1.0$, $\lambda_{cons} = 1.0$, and $\lambda_{bd} = 0.5$. The project implementation does not use RandAugment for GeoRisk-SPC. The student branch receives the project’s `_add_noise` perturbation: clipped additive Gaussian noise with standard deviation 0.1 and clipping range $[-0.2, 0.2]$ when `strong=s`. The teacher branch uses the unperturbed unlabeled image under this setting.

We compare two types of baselines. “Reimpl.” results are reproduced in our codebase using the same UNet backbone, data splits, and training schedule, enabling a controlled comparison. “Official” results use the authors’ released code with its own architecture and augmentation choices; they are included for reference but are not directly comparable in backbone capacity or training protocol.

C. Main Results

Table I reports test Dice for all methods. GeoRisk-SPC-DG reaches 88.26%, 90.42%, 90.90%, and 91.41% at 5%, 10%, 20%, and 40% labels. It is the strongest method at 10%, 20%, and 40% labels, and ties SegMatch (reimpl.) at 5%. Compared with UniMatch (reimpl.), GeoRisk-SPC-DG improves by +0.90, +2.24, +1.04, and +2.19 points. Compared with SegMatch (reimpl.), it changes by +0.00, +1.33, +0.73, and +1.42 points. The 10% GeoRisk-SPC-DG result (90.42%) also exceeds the 40% UniMatch (reimpl.) result (89.22%) and the 40% SegMatch (reimpl.) result (89.99%), indicating improved label efficiency under this protocol.

The depth-guided encoder is the clearest driver. On the Mean Teacher baseline, adding DepthGuiderV4 improves 20% test Dice from 87.43% to 88.83% (+1.40). Within our method, GeoRisk-SPC-DG improves over GeoRisk-SPC-UNet by +1.58, +1.83, +2.18, and +0.26 points across the four label ratios. This supports the encoder as the most stable empirical contribution.

D. Multi-class Generalization

To evaluate whether the geometric risk principle extends beyond binary segmentation, we test all methods on EndoVis 2017 Task 2 (4-class instrument part segmentation: shaft, wrist, clasper, and probe) and Task 3 (7-class instrument type segmentation). Tables II and III report results across all label ratios.

On Task 2 (Table II), GeoRisk-SPC-DG achieves the best or second-best result at every label ratio: 61.86% at 5%, 65.56% at 10%, 65.41% at 20%, and 67.56% at 40%. At 5% labels it leads by +3.26 over MT, and at 40% it trails URPC by only 0.14 points. On Task 3 (Table III), GeoRisk-SPC-DG reaches 20.13% at 20% labels, competitive with SegMatch (24.02%) and URPC (24.07%) given the much harder 7-class setting. U2PL suffers from NaN loss during training, producing near-zero Dice (3–7%), which suggests that its contrastive prototype learning may be unstable under extreme class imbalance in the 7-class setting. These results indicate that depth-guided representation learning generalizes to multi-class surgical segmentation, while the risk supervision module may need class-specific threshold calibration for finer-grained tasks.

E. Ablation Studies

Table IV reports 20% label-ratio test Dice for component and threshold variants. The ablation is intentionally not monotonic. The full default configuration reaches 90.96% Dice, but several variants are higher: Depth+Uncertainty reaches 91.49%, removing risk localization reaches 91.42%, and using the DG encoder without risk supervision reaches 91.29%. This means the default fixed-threshold risk partition is not yet optimal.

The main diagnosis is threshold calibration. With $\tau_c = 0.9$, the low-risk hard pseudo-label stream is dominated by easy background, while useful foreground and boundary supervision becomes narrow. In training-batch diagnostics, the effective non-background low-risk signal can shrink to the 0.2–1.8% pixel range after excluding trivial background. In this condition, $\mathcal{L}_{pl}$ provides insufficient hard foreground grounding, and the high-risk consistency and boundary terms can dominate the nontrivial regions. Removing risk localization assigns hard pseudo-labels to all pixels, which is less selective but supplies a stronger hard-label training signal; this explains why “w/o risk localization” improves Dice by +0.46 over the default.

The ablation still supports the geometric premise. Depth+Uncertainty is the best variant at 91.49% (+0.53), indicating that depth discontinuity paired with uncertainty is an effective risk prior. Removing $\mathcal{L}_{bd}$ gives the largest negative component effect, 90.39% (−0.57), confirming that boundary consistency is valuable. The threshold variants also show that fixed thresholds matter: $\tau_c = 0.95$ drops to 90.43%, and $\tau_r = 0.3$ drops to 90.64%. The correct conclusion is therefore not that every risk component monotonically improves Dice, but that geometric risk is useful and must be calibrated so that hard pseudo-label supervision remains sufficiently grounded.

TABLE I
TEST DICE ON ENDOVIS 2017 WITH 4-FOLD CROSS-VALIDATION. VALUES ARE MEAN $\pm$ STD OVER FOLDS.

Method	5%	10%	20%	40%
Sup-only	—	—	87.51 $\pm$ 3.53	—
MT [4]	84.27 $\pm$ 15.51	85.17 $\pm$ 15.06	87.43 $\pm$ 4.07	86.46 $\pm$ 15.11
MT-DGv4	—	87.35 $\pm$ 15.52	88.83 $\pm$ 4.06	—
UAMT [26]	84.94 $\pm$ 14.82	85.25 $\pm$ 15.16	86.94 $\pm$ 4.39	86.99 $\pm$ 14.37
URPC [27]	85.30 $\pm$ 15.34	86.79 $\pm$ 14.77	87.87 $\pm$ 4.31	87.17 $\pm$ 14.39
CPS [6]	85.39 $\pm$ 12.45	81.38 $\pm$ 13.31	89.88 $\pm$ 9.82	87.92 $\pm$ 11.19
MMS [29]	85.75 $\pm$ 12.14	87.44 $\pm$ 11.46	87.66 $\pm$ 11.63	84.35 $\pm$ 12.72
UniMatch (reimpl.) [8]	87.36 $\pm$ 12.29	88.18 $\pm$ 12.24	89.86 $\pm$ 9.27	89.22 $\pm$ 11.06
UniMatch (official) [8]	85.29 $\pm$ 13.82	86.16 $\pm$ 12.39	86.61 $\pm$ 10.73	88.87 $\pm$ 11.15
SegMatch (reimpl.) [18]	88.26 $\pm$ 11.50	89.09 $\pm$ 10.52	90.17 $\pm$ 9.20	89.99 $\pm$ 10.17
SegMatch (official) [18]	86.42 $\pm$ 13.22	86.39 $\pm$ 4.21	81.82 $\pm$ 7.77	86.39 $\pm$ 4.21
U2PL (official) [7]	86.14 $\pm$ 4.45	87.98 $\pm$ 4.77	88.59 $\pm$ 4.22	89.57 $\pm$ 4.18
CorrMatch (official) [9]	85.45 $\pm$ 5.78	86.22 $\pm$ 5.41	87.87 $\pm$ 4.89	89.07 $\pm$ 3.88
CW-BASS (official) [10]	85.48 $\pm$ 5.61	—	—	—
GeoRisk-SPC (UNet)	86.68 $\pm$ 15.81	88.59 $\pm$ 13.27	88.72 $\pm$ 12.99	91.15 $\pm$ 8.91
GeoRisk-SPC-DG	88.26 $\pm$ 12.75	90.42 $\pm$ 10.71	90.90 $\pm$ 10.26	91.41 $\pm$ 10.29

TABLE II
TEST DICE ON ENDOVIS 2017 TASK 2 (4-CLASS PART SEGMENTATION) WITH 4-FOLD CROSS-VALIDATION. VALUES ARE FOLD-LEVEL MEAN $\pm$ STD.

Method	5%	10%	20%	40%
Fully	58.63 $\pm$ 7.98	61.21 $\pm$ 8.80	64.62 $\pm$ 8.15	66.54 $\pm$ 9.79
MT [4]	58.60 $\pm$ 8.61	62.61 $\pm$ 7.84	63.03 $\pm$ 8.49	66.17 $\pm$ 9.28
MT-DGv4	61.42 $\pm$ 9.63	64.09 $\pm$ 7.28	66.18 $\pm$ 9.07	66.53 $\pm$ 11.88
UAMT [26]	59.05 $\pm$ 8.86	63.50 $\pm$ 9.55	64.22 $\pm$ 7.13	67.34 $\pm$ 9.61
URPC [27]	61.05 $\pm$ 9.70	65.60$\pm$8.19	67.27$\pm$8.35	67.70$\pm$10.57
CPS [6]	58.60 $\pm$ 7.88	61.85 $\pm$ 8.23	63.49 $\pm$ 7.97	66.20 $\pm$ 10.06
UniMatch [8]	58.07 $\pm$ 8.67	61.90 $\pm$ 10.06	63.54 $\pm$ 10.44	66.14 $\pm$ 11.56
SegMatch [18]	60.15 $\pm$ 8.92	62.86 $\pm$ 7.58	64.88 $\pm$ 9.42	66.04 $\pm$ 8.19
U2PL [7]	57.43 $\pm$ 7.77	58.15 $\pm$ 6.03	58.75 $\pm$ 5.30	57.61 $\pm$ 6.51
CorrMatch [9]	55.72 $\pm$ 8.28	61.57 $\pm$ 9.31	64.03 $\pm$ 9.05	64.62 $\pm$ 9.65
CW-BASS [10]	58.26 $\pm$ 8.68	61.40 $\pm$ 8.16	64.31 $\pm$ 8.51	66.40 $\pm$ 9.04
GeoRisk-SPC-DG	61.86$\pm$8.49	65.56 $\pm$ 7.31	65.41 $\pm$ 7.07	67.56 $\pm$ 7.49

TABLE III
TEST DICE ON ENDOVIS 2017 TASK 3 (7-CLASS TYPE SEGMENTATION) WITH 4-FOLD CROSS-VALIDATION. VALUES ARE FOLD-LEVEL MEAN $\pm$ STD.
U2PL SUFFERS FROM NAN LOSS DURING TRAINING, PRODUCING NEAR-ZERO DICE.

Method	5%	10%	20%	40%
Fully	14.85 $\pm$ 5.88	21.00 $\pm$ 10.49	20.75 $\pm$ 10.09	21.11 $\pm$ 7.64
MT [4]	15.24 $\pm$ 5.58	19.10 $\pm$ 7.10	17.59 $\pm$ 7.77	23.65 $\pm$ 6.73
MT-DGv4	19.53 $\pm$ 6.64	19.95 $\pm$ 6.18	21.16 $\pm$ 9.95	23.08 $\pm$ 10.11
UAMT [26]	15.26 $\pm$ 5.92	19.79 $\pm$ 6.78	20.82 $\pm$ 6.78	24.35 $\pm$ 5.92
URPC [27]	19.66$\pm$3.43	25.11$\pm$12.18	24.07$\pm$10.22	29.64$\pm$11.38
CPS [6]	18.66 $\pm$ 4.98	20.40 $\pm$ 7.54	19.26 $\pm$ 6.72	22.60 $\pm$ 5.94
UniMatch [8]	14.67 $\pm$ 6.38	21.12 $\pm$ 8.12	21.48 $\pm$ 6.32	21.77 $\pm$ 5.23
SegMatch [18]	18.24 $\pm$ 4.82	20.23 $\pm$ 8.54	24.02 $\pm$ 10.38	24.58 $\pm$ 4.08
U2PL [7]	3.46 $\pm$ 1.24	6.96 $\pm$ 7.45	6.75 $\pm$ 6.78	8.84 $\pm$ 4.98
CorrMatch [9]	15.36 $\pm$ 7.03	20.50 $\pm$ 9.11	20.72 $\pm$ 8.30	21.49 $\pm$ 8.60
GeoRisk-SPC-DG	19.40 $\pm$ 11.19	21.13 $\pm$ 7.20	20.13 $\pm$ 11.33	21.39 $\pm$ 8.91

F. Risk Localization Diagnostics

We audit the default masks on the 20% GeoRisk-SPC-DG setting using the converged fold checkpoints as teacher proxies. Table V reports pooled 4-fold statistics on held-out test frames. The all-pixel M_l coverage is high because easy background dominates binary instrument segmentation. This number alone is misleading. The important behavior is the accuracy gap: pseudo-label accuracy is 98.72% inside M_l but only 63.52% outside M_l , and foreground pseudo-label accuracy is 95.97% inside low-risk foreground pixels but 68.53% outside them. Thus the risk partition separates reliable and unreliable pseudo-labels, but the default thresholds leave the nontrivial boundary/high-risk set small. This supports the

ablation diagnosis: risk localization is directionally correct, while its fixed thresholds are not yet calibrated for optimal hard-label signal.

G. Boundary Quality Metrics

Table VI reports fold-level boundary-aware metrics at 20% labels. GeoRisk-SPC-DG leads all baselines in every metric. The largest improvement is Boundary F1: 91.11% versus 86.19% for MT (+4.92), 86.66% for SegMatch (+4.45), and 86.38% for UAMT (+4.73). HD95 drops from 19.3–20.8 to 14.9, and ASSD drops from 3.16–3.30 to 2.45. These results indicate that the depth-guided encoder and risk-guided

TABLE IV
ABLATION RESULTS AT 20% LABELS. VALUES ARE TEST DICE.

Configuration	Test Dice	Δ vs Full
Depth + Uncertainty	91.49 $\pm$ 3.53	+0.53
w/o risk localization	91.42 $\pm$ 3.41	+0.46
DG encoder w/o risk	91.29 $\pm$ 3.62	+0.33
w/o $\mathcal{L}_{cons}$	91.25 $\pm$ 3.36	+0.29
$\tau_r = 0.7$	91.23 $\pm$ 3.41	+0.27
Random mask ($\tau_r = 1.0$)	91.19 $\pm$ 3.46	+0.23
Depth only	91.13 $\pm$ 3.27	+0.17
Conflict only	91.12 $\pm$ 3.39	+0.16
$\tau_c = 0.7$	91.07 $\pm$ 3.40	+0.11
Uncertainty only	90.98 $\pm$ 3.91	+0.02
Full method	90.96 $\pm$ 3.86	—
$\tau_r = 0.3, \tau_c = 0.7$	90.66 $\pm$ 3.89	-0.30
$\tau_r = 0.3$	90.64 $\pm$ 4.92	-0.32
$\tau_c = 0.95$	90.43 $\pm$ 4.07	-0.53
w/o $\mathcal{L}_{bd}$	90.39 $\pm$ 4.78	-0.57

TABLE V
POST-HOC RISK-MASK AUDIT FOR 20% GEORISK-SPC-DG ON HELD-OUT TEST FRAMES. COVERAGE IS THE POOLED PERCENTAGE OF IMAGE PIXELS.

Region	Coverage	Pseudo-label accuracy
Low-risk M_l	97.94%	98.72%
High-risk M_r	0.58%	58.41%
Outside M_l	2.06%	63.52%
Low-risk foreground	11.28%	95.97%
Outside- M_l foreground	1.04%	68.53%

supervision improve boundary precision, not only region-level Dice.

H. Depth-guided Encoder Analysis

The architecture comparison gives the most stable positive result. MT-DGv4 improves over MT by +1.40 points at 20% labels. GeoRisk-SPC-DG improves over GeoRisk-SPC-UNet at every label ratio, with the largest gains at 5%, 10%, and 20%. This pattern indicates that depth is useful both as a feature source and as a risk prior, but the feature-source contribution is currently more robust than the fixed-threshold risk-supervision contribution.

Figure 1 provides an illustrative diagnostic of depth feature activation across encoder levels. It should be interpreted as a qualitative analysis aid, not as quantitative evidence. The observed pattern is consistent with the design motivation: shallow levels preserve boundary-sensitive geometric information, while deeper levels can suppress depth residuals when semantic RGB features are sufficient.

I. Qualitative Analysis

Figure 2 shows a real held-out test example from the 20% GeoRisk-SPC-DG fold. The 8-panel visualization includes input image, pre-computed depth map, teacher uncertainty, depth discontinuity, geometry-semantic conflict, risk map with M_r overlay, pseudo-label error map, and prediction overlay with GT boundary. High-risk responses (yellow contour in panel f) concentrate near instrument boundaries and depth transitions, consistent with the intended risk localization behavior. The pseudo-label error map (panel g) shows that teacher errors

TABLE VI
BOUNDARY-AWARE METRICS AT 20% LABELS (FOLD-LEVEL MEAN $\pm$ STD).

Method	Dice	BF1	HD95	ASSD
MT	88.48 $\pm$ 5.63	86.19 $\pm$ 7.30	20.81 $\pm$ 10.98	3.29 $\pm$ 1.68
SegMatch	89.30 $\pm$ 5.63	86.66 $\pm$ 8.16	20.09 $\pm$ 14.65	3.30 $\pm$ 2.34
UAMT	88.55 $\pm$ 5.40	86.38 $\pm$ 7.09	19.33 $\pm$ 12.68	3.16 $\pm$ 1.98
GeoRisk-SPC-DG	91.49$\pm$4.44	91.11$\pm$5.35	14.93$\pm$10.85	2.45$\pm$1.83

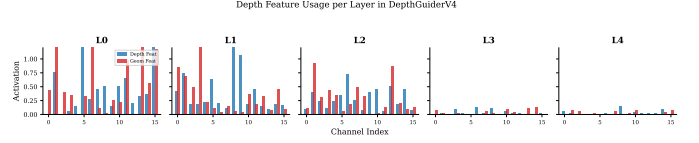


Fig. 1. Illustrative diagnostic of DepthGuiderV4 feature usage across encoder levels. Real per-fold qualitative visualizations will be added in the final visualization package; this figure is not used as quantitative evidence.

cluster in similar regions, supporting the geometric risk hypothesis. Additional examples from other held-out sequences are provided in the supplementary material.

V. DISCUSSION

A. What the Results Support

The experiments support four claims. First, depth-guided representation learning is valuable for surgical instrument segmentation under limited labels. The MT-to-MT-DG and GeoRisk-SPC-UNet-to-GeoRisk-SPC-DG comparisons show consistent gains. Second, depth is a meaningful pseudo-label risk prior. Depth+Uncertainty gives the best ablation result, and the mask audit shows a large accuracy gap between low-risk and outside-low-risk foreground predictions. Third, boundary quality improves alongside region-level Dice. GeoRisk-SPC-DG achieves Boundary F1 of 91.11% versus 86.19% for MT, and reduces HD95 from 20.8 to 14.9 pixels. This indicates that the depth-guided encoder and risk-guided supervision improve boundary precision, not only region overlap. Fourth, fixed-threshold risk localization is not yet sufficient. The default $\tau_c = 0.9$ and $\tau_r = 0.5$ setting can under-supply hard foreground pseudo-label grounding in difficult regions, producing the observed non-monotonic ablation table.

The multi-class experiments (Tables II–III) provide additional context. On Task 2 (4-class part segmentation), GeoRisk-SPC-DG remains competitive (65.41% at 20% labels, within 1.86 points of URPC). On Task 3 (7-class type segmentation), all methods suffer significant Dice drops due to class imbalance and finer semantic granularity, with GeoRisk-SPC-DG at 20.13%. This suggests that the risk supervision module’s fixed thresholds may need class-specific calibration when the label space expands.

B. Limitations

The main limitation is threshold calibration. GeoRisk-SPC currently uses fixed τ_r and τ_c , and the ablation shows that this can make risk localization suboptimal. Adaptive thresholds, class-balanced pseudo-label coverage, or curriculum schedules should replace the fixed default. A second limitation is

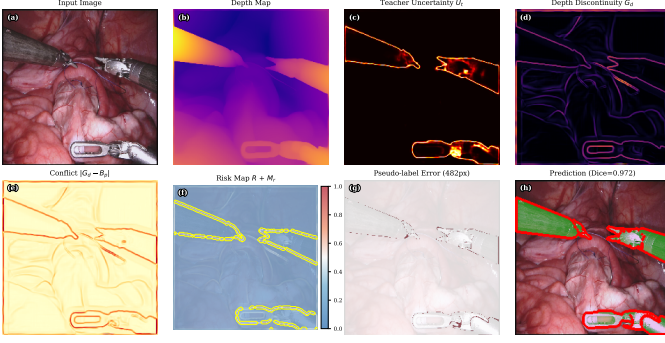


Fig. 2. Real held-out test example from 20% GeoRisk-SPC-DG. (a) Input image. (b) Pre-computed depth map. (c) Teacher uncertainty U_t . (d) Depth discontinuity G_d . (e) Geometry-semantic conflict $|G_d - B_p|$. (f) Risk map R with M_r boundary (yellow). (g) Pseudo-label error (red). (h) Prediction overlay (green) with GT boundary (red). Best viewed in color.

depth quality. Monocular depth may be inaccurate in specular, smoke-filled, or textureless surgical frames, and the risk map inherits these errors. Third, while we now report results on Task 2 (4-class) and Task 3 (7-class) in addition to Task 1 (binary), the multi-class performance gap is significant. On Task 3, GeoRisk-SPC-DG drops to 20.13% Dice at 20% labels, suggesting that class-specific risk calibration and foreground-aware thresholding are needed for finer-grained segmentation. Cross-dataset generalization and additional surgical procedures remain untested. Fourth, with only 4 cross-validation folds, Wilcoxon signed-rank tests cannot reach the $p < 0.05$ threshold (minimum achievable $p = 0.0625$ for $n = 4$). The improvements are consistent in direction but lack formal statistical significance under this fold count.

VI. CONCLUSION

We presented GeoRisk-SPC, a geometry-conditioned pseudo-label risk supervision framework for semi-supervised surgical instrument segmentation. The core idea is to use depth discontinuity not merely as an additional segmentation input, but as a cue for deciding where pseudo-labels should not be trusted. The full GeoRisk-SPC-DG model achieves test Dice scores of 88.26%, 90.42%, 90.90%, and 91.41% at 5%, 10%, 20%, and 40% labels on EndoVis 2017. The strongest empirical contribution is DepthGuiderV4, while the risk-supervision results reveal an important limitation: fixed confidence and risk thresholds can make hard foreground pseudo-label supervision too sparse. This makes GeoRisk-SPC a promising but not yet fully calibrated direction. Future work should learn adaptive risk thresholds, balance foreground/background pseudo-label coverage, and validate the method on multi-class and cross-dataset surgical segmentation benchmarks.

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